

ARISE: Evidence-Grounded Aspect-Based Sentiment Analysis of Student Feedback on College Mental Health Support

Tiansui Xie, Palak Sood, Divyanshu Gupta, Sang Won Bae, Yue Ning, Ping Wang
Stevens Institute of Technology, Hoboken, NJ, USA

{txie13, sbae4, yning5, pwang44}@stevens.edu, palaksood1997@gmail.com, divyanshu30gupta@gmail.com

Abstract—Mental health support plays an important role in college students’ well-being, yet understanding student perceptions remains challenging due to the complex and multi-dimensional nature of open-ended feedback. Prior work has largely focused on document-level sentiment analysis, which captures overall polarity but fails to reveal fine-grained opinions across specific service aspects. In this work, we construct AWARE-SMILE, a fine-grained extension of SMILE-College dataset with a refined eight-category aspect taxonomy, aspect-level sentiment labels, and supporting evidence spans. We propose ARISE (Aspect Reasoning with Iterative Sentiment Enhancement), a two-stage framework that decomposes the task into aspect category detection (ACD) and aspect sentiment classification (ASC). We introduce an iterative propose-to-verify paradigm, where predictions are validated against evidence grounded in the input text and refined through feedback. We evaluate ARISE against traditional machine learning models, fine-tuned transformers, and large language models, showing improved performance in both aspect detection and sentiment classification, along with stronger evidence grounding and interpretability, supported by qualitative analyses of model behavior. Our work provides a scalable, evidence-grounded approach for helping institutions identify service gaps, understand student needs, and improve mental health support in higher education.

Index Terms—Aspect-based sentiment analysis, large language models, mental health, student feedback, retrieval-augmented generation, iterative reasoning

I. INTRODUCTION

The mental well-being of college students has become an increasingly important concern, as rising rates of stress, anxiety, and depression continue to affect students’ academic performance, social relationships, and long-term outcomes [1], [2]. Recent national reports further highlight the growing demand for campus mental health support services [3]. In response, many institutions have expanded counseling centers, therapy programs, peer support initiatives, and crisis hotlines. However, students still frequently report concerns regarding counseling quality, responsiveness, and service accessibility, making it important to better understand student perceptions toward different components of campus mental health services.

Recent studies have increasingly explored NLP-based approaches for analyzing open-ended student feedback. Most existing work relies on document-level sentiment classification, where each response is assigned a single overall sentiment label [4]. Although transformer-based models and large language models (LLMs) [5]–[7] have achieved strong perfor-

TABLE I
REPRESENTATIVE ASPECT-SENTIMENT EXAMPLES FROM
AWARE-SMILE DATASET.

Student Response Spans	Aspect	Sentiment
Focus on taking care of yourself has been emphasized.	Wellness Service	Positive
my college works well in communicating about the various mental health resources on campus.	On-campus Service	Positive
Reaching out to the students, not a lot of people are aware of the facilities provided by the university.	Service Availability	Neutral
Understanding what happens if you do need to get help and what comes next is lacking.	Therapy Service	Negative
teachers are not understanding even when students have accommodations.	General	Negative

mance in capturing global sentiment patterns, document-level sentiment analysis assumes sentiment homogeneity within a response. This assumption is often violated in mental health feedback, where students may express both positive and negative opinions toward different service components. For example, a student may praise counseling quality while criticizing appointment availability or institutional responsiveness. As a result, aggregate sentiment labels provide limited insight into which specific service dimensions drive student experiences.

To address this limitation, Aspect-Based Sentiment Analysis (ABSA) provides a more fine-grained framework for identifying service-related aspects together with their associated sentiment polarity [8]–[10]. Such fine-grained modeling is particularly important in campus mental health settings, where institutional interventions often target specific operational issues rather than overall sentiment trends.

The SMILE-College dataset [11] was introduced to support research on student experiences with campus mental health services through open-ended survey responses. Existing studies on SMILE-College mainly focus on document-level sentiment classification using LLMs. However, these approaches overlook fine-grained aspect-level structures and do not explicitly verify whether generated predictions are reliably grounded in the input text. Recent studies further show that prompting strategies such as zero-shot reasoning and self-refinement still underperform task-specific ABSA systems in

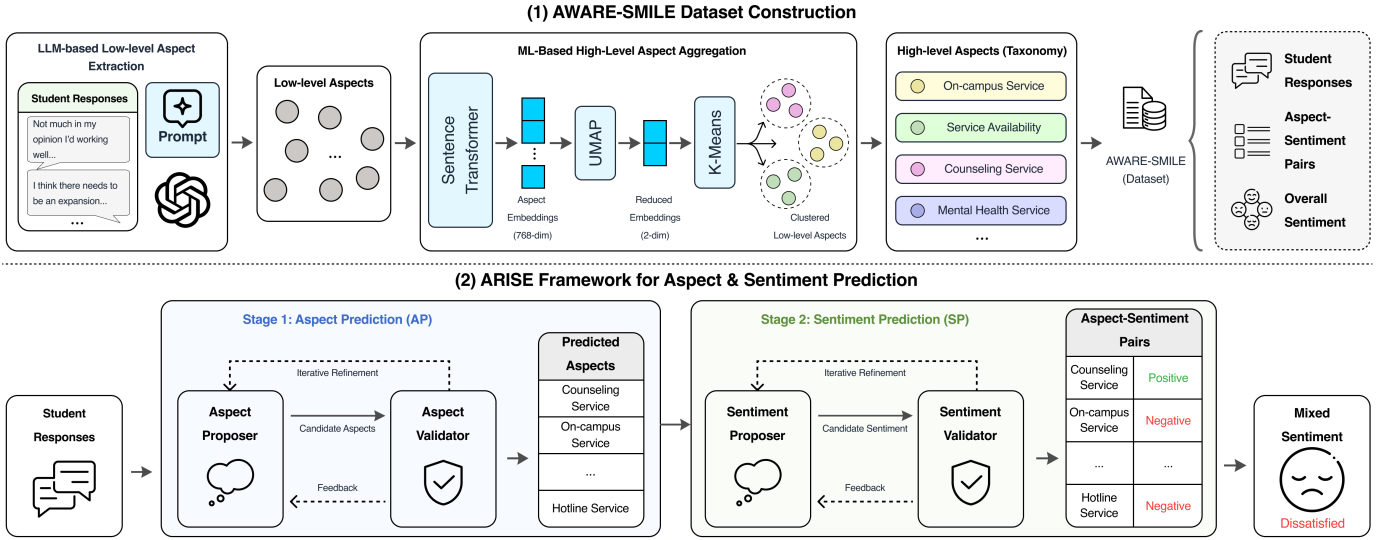


Fig. 1. Overview of the High-level aspect extraction pipeline, AWARE-SMILE structure, and ARISE framework, including LLM-based aspect annotation, ML-driven aspect aggregation, and iterative aspect-sentiment prediction via Aspect Predictor (AP) and Sentiment Predictor (SP).

structured sentiment extraction settings, and iterative prompting methods may also suffer from reasoning inconsistency and error accumulation during multi-turn refinement [12]. These limitations suggest that while LLMs provide strong generalization capability, additional validation mechanisms are needed to improve prediction reliability in fine-grained aspect-level sentiment analysis.

Motivated by these challenges, this work extends the analysis of SMILE-College through a structured ABSA framework tailored to college mental health feedback. We propose ARISE, a verification-oriented proposer-validator framework that integrates iterative reasoning with retrieval-grounded validation for both aspect prediction and sentiment prediction. By combining LLM-based generation with memory-based verification, ARISE improves prediction reliability without updating the parameters of the LLM backbones. Our main contributions are summarized as follows.

- We construct **AWARE-SMILE** (Aspect-Wise Annotation and Response Evaluation on SMILE-College), a fine-grained aspect-based sentiment dataset built upon SMILE-College, containing manually annotated aspect categories, sentiment polarities, and evidence spans for student feedback on campus mental health services. We further develop a refined aspect taxonomy through LLM-assisted extraction and clustering. Table I presents representative examples from AWARE-SMILE, which cover student response spans with different aspects and their corresponding sentiments. The public repository for code, dataset, and supplementary materials is available at this link¹.
- We propose **ARISE** (Aspect Reasoning with Iterative Sentiment Enhancement), a two-stage proposer-validator framework for ABSA task, where retrieval-augmented validation and iterative refinement are introduced to im-

prove reliability and interpretability. Figure 1 presents an overview of the proposed framework and dataset construction pipeline.

- We conduct experiments across traditional machine learning models, transformer baselines, task-specific ABSA models, and multiple LLM backbones, demonstrating consistent improvements in both aspect prediction and sentiment classification, together with additional analyses on ablation settings, iterative refinement behavior, and case study.

II. RELATED WORK

Sentiment Analysis and Aspect-Based Opinion Mining.

Sentiment analysis is a well-established NLP task that aims to identify the polarity or emotional tone expressed in textual data. Traditional document-level sentiment analysis assigns a single sentiment label to an entire text and has been widely applied to product reviews, social media posts, and educational feedback [13]–[16]. However, such approaches provide only an aggregate polarity judgment and cannot capture fine-grained opinions toward specific entities, attributes, or service components mentioned within the same text. Aspect-based sentiment analysis (ABSA) addresses this limitation by linking sentiment polarity to specific aspects discussed in the text. Existing ABSA studies commonly decompose the task into subtasks such as Aspect Term Extraction (ATE), Aspect Category Detection (ACD), and Aspect Sentiment Classification (ASC) [8]. Benchmark efforts such as SemEval-2014 Task 4, SemEval-2015 Task 12, and SemEval-2016 Task 5 [17]–[19] have established standardized evaluation settings for ABSA and driven the development of methods ranging from rule-based systems and classical machine learning models to recent transformer-based neural approaches. Recent studies have also explored unsupervised and self-supervised aspect discovery to reduce reliance on costly manual annotation, including

¹<https://github.com/voId07/arise-absa>

contrastive learning and attention-based mapping strategies for aspect representation learning [20]–[22].

LLMs for Aspect-Level Sentiment Analysis. Recent studies have increasingly explored prompt-based and few-shot approaches for sentiment analysis and ABSA using large language models (LLMs) [23]. Recent LLM-based ABSA studies have also incorporated transfer learning, iterative self-reflection, and reasoning mechanisms to improve prediction consistency and reduce hallucinations [24], [25]. More comprehensive formulations, such as Aspect–Sentiment Quadruple Prediction (ASQP), aim to jointly extract aspect terms, aspect categories, opinion expressions, and sentiment polarities, enabling richer representations of opinion structures [10], [26]. In parallel, recent work has also explored continual-learning-based and retrieval-enhanced ABSA frameworks for improving cross-domain generalization and structured sentiment reasoning [27]. However, recent empirical analyses suggest that prompting strategies such as zero-shot reasoning, self-improvement, and self-debate still underperform fine-tuned task-specific models in structured ABSA settings, and existing iterative prompting methods may also suffer from reasoning inconsistency and error accumulation during multi-turn refinement, particularly in fine-grained structured extraction tasks with complex output requirements [12], [28]. These limitations motivate validation-enhanced frameworks that explicitly verify aspect and sentiment predictions against textual evidence.

Mental Health Feedback in Higher Education. NLP and text mining methods have also been applied to study student mental health and well-being in educational contexts. Prior work has analyzed textual data from surveys and social media to identify patterns related to stress, anxiety, depression, and help-seeking behaviors among college students and broader populations using NLP and computational social science methods [29]–[32]. The SMILE-College corpus [11] provides a domain-specific dataset of open-ended student responses about campus mental health services. Existing analysis on this dataset mainly focuses on overall sentiment classification, which is useful for estimating general attitudes but does not reveal which service components contribute to positive or negative student experiences. Our work extends this line of research by constructing **AWARE-SMILE**, a span-level aspect–sentiment dataset derived from SMILE-College, and by proposing **ARISE**, a validation-enhanced ABSA framework tailored to higher-education mental health service feedback.

III. AWARE-SMILE DATASET CONSTRUCTION

This study uses the SMILE-College dataset [11], a publicly available corpus of open-ended student feedback regarding campus mental health and wellness services. The dataset contains 794 text responses collected from college students, where each response is associated with one overall sentiment label. The original dataset includes four sentiment categories: *dissatisfied* (377 samples), *mixed* (220 samples), *satisfied* (107 samples), and *neutral* (90 samples). Building upon SMILE-College, we construct **AWARE-SMILE**, a fine-grained aspect-based sentiment dataset that extends the original document-

level annotations with aspect categories, aspect-level sentiment labels, and supporting evidence spans for each response.

We designed a three-step annotation procedure. First, we identified the aspect labels to be used for annotation by extracting candidate aspects from the full SMILE-College dataset and aggregating them into high-level categories. Second, annotators labeled the relevant aspects and their corresponding sentiment polarities, along with the original mentions, as supporting evidence for each response. Finally, we consolidated the annotations through majority voting to produce the final aspect and sentiment labels.

A. Aspect Taxonomy Construction

Building on SMILE-College, we identified service-related aspects mentioned across student responses through a two-stage process, including extracting candidate aspect terms from all student responses and clustering semantically related terms into broader aspect categories. Specifically, in the first stage, shown as **AWARE-SMILE Dataset Construction** in Figure 1, we first used GPT-4o mini to extract up to three salient service-related terms per response (temperature 0.1; top- p 0.5). The prompt included the task, survey question, response, and previously extracted terms. These terms formed a dynamic reference list that encouraged reuse of an existing semantically equivalent term before creating a new one. This extraction step produced 184 coarse-grained candidate aspects, which we refer to as **low-level aspects**.

To obtain a more coherent and fine-grained aspect taxonomy, we encoded the 184 terms and projected them to two dimensions using cosine-distance UMAP, and selected 13 K-means clusters by the highest silhouette score over $k = 2, \dots, 20$. Manual review then merged overlapping groups and reassigned misplaced terms using service-domain definitions, without using sentiment labels. This produced eight **high-level aspects**: *On-campus Service*, *Service Availability*, *Counseling Service*, *Mental Health Service*, *Wellness Service*, *Therapy Service*, *Hotline Service*, and *General*.

B. Aspect and Sentiment Annotation

Based on the eight high-level aspects identified above, three research-trained annotators independently labeled the aspect categories and corresponding sentiment polarity for each data sample. Annotators followed a documented labeling guide that defines each aspect category and provides representative examples and decision rules. For each data sample, annotators first identified the relevant aspect spans and then assigned a sentiment polarity to each labeled aspect. We consider three aspect-level sentiment labels: *positive* (*pos*), *negative* (*neg*), and *neutral* (*neu*), corresponding to supportive or favorable opinions, critical or unfavorable opinions, and non-evaluative or ambiguous statements, respectively. To improve annotation consistency, we included an additional sentiment consistency verification step. Since each data sample in SMILE-College already contains an overall document-level sentiment label, we aggregated the annotated aspect-level sentiments to infer an alternative overall sentiment and compared it with the

TABLE II
BASIC STATISTICS OF THE AWARE-SMILE DATASET.

Statistic	Value
Avg. # Aspects per Sample	1.316
Min # Aspects per Sample	1
Max # Aspects per Sample	3
Samples with Multiple Aspects (%)	29.5%
Most Frequent Aspect	Service Availability (NEG) - 280
Least Frequent Aspect	Mental Health Service (NEU) - 2
Positive Aspect-Sentiment Pairs	359
Neutral Aspect-Sentiment Pairs	107
Negative Aspect-Sentiment Pairs	653
Avg. # Spans per Sample	1.409
Min # Spans per Sample	1
Max # Spans per Sample	5
Avg. Evidence Span Length (words)	16.100
Min Evidence Span Length (words)	1
Max Evidence Span Length (words)	78

original dataset label. When inconsistencies were identified, the corresponding samples underwent additional review and adjudication to obtain the final annotations.

C. Consensus Label Finalization

After annotation, we consolidated the collected labels through fine-grained span-level majority voting. Since free-form spans may differ slightly at their boundaries while referring to the same content, we aligned spans using an Intersection-over-Union (IoU) threshold of 0.6. This threshold requires substantial character-level overlap while tolerating minor boundary differences, and was fixed for all samples and annotators before label aggregation. For each resulting span group, the final annotations for aspect categories and sentiment labels were obtained by majority voting among the three annotators. Specifically, labels supported by at least two annotators were accepted as the final annotation. Cases without an initial majority agreement underwent additional review and adjudication to determine the final labels. Before majority voting and adjudication, we measured agreement on the three original annotation sets, normalizing *suggestion* and *limitation* labels to *negative*. Fleiss’ κ , pooled over the eight binary aspect-presence decisions, was 0.460. One-to-one pairwise character-span alignment at $\text{IoU} \geq 0.6$ yielded a mean evidence-span F1 of 0.749; conditional on aligned spans with matching aspect labels, the mean pairwise Cohen’s κ for sentiment was 0.929. This process produced **AWARE-SMILE**, which contains 794 annotated samples with one or more high-level aspect labels and their corresponding sentiment polarities for each data sample. Table I presents representative examples from AWARE-SMILE, and Table II summarizes the overall dataset statistics, including aspect distributions, aspect-sentiment pairs, and evidence span characteristics.

IV. THE ARISE FRAMEWORK

This section introduces ARISE, a two-stage iterative verification framework for aspect-based sentiment analysis of open-ended student feedback. Given an input comment, ARISE first identifies the service-related aspects discussed in the text

and then predicts the sentiment polarity express toward each detected aspect. ARISE consists of two modules: the Aspect Predictor (AP) and the Sentiment Predictor (SP), illustrated in Figure 2 and Figure 3, respectively. Both modules follow a unified propose-to-verify paradigm. A proposer first generates an initial prediction, and a validator then checks whether the prediction is supported by evidence grounded in the input text and by relevant examples. Predictions with insufficient support are iteratively refined through validator feedback.

A. Aspect Predictor

Given an input comment $x = \{w_1, w_2, \dots, w_n\}$ and a predefined aspect schema $\mathcal{C} = \{c_1, c_2, \dots, c_K\}$, the Aspect Predictor (AP) identifies the aspect categories that are explicitly discussed in the input text. The **Aspect Proposer** first generates an initial set of candidate aspects:

$$C^{(0)} = \{(a_i, E_{a_i})\}_{i=1}^m,$$

where $a_i \in \mathcal{C}$ denotes a candidate aspect and E_{a_i} represents the corresponding evidence spans extracted from the input text. Each evidence span $e \in E_{a_i}$ must satisfy $e \subset x$, meaning it must be a contiguous substring of the original text. This constraint ensures that all proposed aspects are grounded in explicit textual evidence rather than inferred from external knowledge.

To strengthen the verification process, AP incorporates a **Retrieval-augmented Memory Bank** represented by:

$$\mathcal{M}_{\text{ap}} = \{(s_j, a_j)\}_{j=1}^N,$$

where s_j is a training-split annotated span with aspect a_j . The bank is frozen for development and test inference: evaluation annotations are never stored or retrieved.

For a proposed candidate aspect a , the **Aspect Validator** first computes a semantic embedding $q = f(x)$, where $f(\cdot)$ denotes the sentence embedding model. It then retrieves semantically similar positive prototypes from the same aspect category and hard negative prototypes from different categories in the memory bank $\mathcal{M}_{\text{ap}}$:

$$P_a = \text{TopM}_{s_j, a_j=a} \text{sim}(q, f(s_j)),$$

$$N_a = \text{TopN}_{s_j, a_j \neq a} \text{sim}(q, f(s_j)),$$

where $\text{sim}(\cdot, \cdot)$ denotes the embedding similarity function. These retrieved prototypes provide contextual evidence that helps the validator distinguish semantically similar aspect categories and evaluate whether the candidate aspect is sufficiently grounded in the input text. For each candidate aspect, the validator then predicts a decision tuple:

$$d_a = (\text{mentioned}_a, S_a, E_a),$$

where S_a denotes the LLM confidence score based on the retrieved positive and hard negative prototypes P_a and N_a , and $\text{mentioned}_a \in \{0, 1\}$ indicates whether the evidence spans E_a proposed by the aspect proposer are strictly derived from the input text. Each evidence span e must correspond to a substring in the original text with character boundaries $\text{span}(e) = [i, j]$

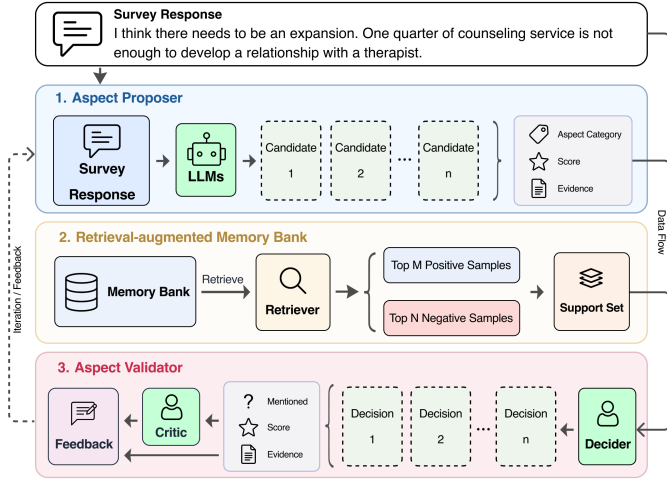


Fig. 2. Overview of the Aspect Predictor module, where an LLM-based proposer generates aspect candidates from survey responses, and a retrieval-augmented validator refines predictions through evidence grounding, positive/negative prototype retrieval, and iterative decision-critique loops.

such that $x[i:j] = e$. If no valid evidence can be located in the text, the validator assigns $\text{mentioned}_a = 0$. This evidence-grounded constraint is designed to reduce hallucinated predictions and ensure that aspect identification remains text-faithful. If the validator assigns $\text{mentioned}_a = 0$, or the confidence score S_a falls below a predefined threshold τ_{ap} , the validator returns critic feedback:

$$\Gamma^{(t)} = \{\Gamma_a^{(t)}\}_{a \in C^{(t)}},$$

where $\Gamma_a = (\text{why_wrong}, \text{confu}, \text{sugg_add}, \text{sugg_drop})$ contains critique information for aspect candidate a , including critiques, potential category confusion, and suggestions for adding or removing aspect categories. The feedback is then used to revise the candidate set:

$$C^{(t+1)} = \text{Revise}(C^{(t)}, \Gamma^{(t)}).$$

This iterative refinement stops when all current candidates pass validation, when the revised candidate set is unchanged from the preceding round, or when the maximum number of iterations T_{ap} is reached. The final output of AP is denoted as:

$$\mathcal{O}_{ap} = \{(a, S_a, E_a)\}_{a \in C^*},$$

where C^* denotes the final validated aspect set after iterative refinement. $\mathcal{O}_{ap}$ contains the detected aspects, their validation scores, and their supporting evidence spans.

B. Sentiment Predictor

The Sentiment Predictor (SP) assigns sentiment polarity to each aspect detected by AP. Given an input comment x and an aspect $a \in \mathcal{C}$, SP predicts an aspect-level sentiment label from the label set $\mathcal{Y} = \{\text{pos}, \text{neg}, \text{neu}\}$, corresponding to the sentiment labels: positive, negative, and neutral. The **Sentiment Proposer** first generates a candidate sentiment:

$$p_a^{(t)} = (y_a^{(t)}, r_a^{(t)}, E_a^{(t)}),$$

Algorithm 1 Aspect Predictor

Require: Text x ; aspect schema $\mathcal{C}$; memory bank $\mathcal{M}_{ap}$; embedding model $f(\cdot)$; threshold τ_{ap} ; max iterations T_{ap}
Ensure: Predicted aspect set C^* and structured output $\mathcal{O}_{ap}$

```

1:  $C^{(0)} \leftarrow \text{ASPECTPROPOSER}(x, \mathcal{C})$ ,  $t \leftarrow 0$ 
2: repeat
3:    $\mathcal{D}^{(t)} \leftarrow \emptyset$ ,  $\Gamma^{(t)} \leftarrow \emptyset$ 
4:   for all  $a \in C^{(t)}$  do
5:      $E_a \leftarrow \text{EXTRACTEVIDENCE}(x, a)$ ;  $q \leftarrow f(x)$ 
6:      $P_a \leftarrow \text{TopM}_{s_j, a_j=a} \text{sim}(q, f(s_j))$ ,  $N_a \leftarrow \text{TopN}_{s_j, a_j \neq a} \text{sim}(q, f(s_j))$ 
7:      $d_a = (\text{mentioned}_a, S_a, E_a) \leftarrow \text{ASPECTVALIDATOR}(x, a, E_a, P_a, N_a)$ 
8:      $\mathcal{D}^{(t)} \leftarrow \mathcal{D}^{(t)} \cup \{d_a\}$ 
9:     if  $\text{mentioned}_a = 0$  or  $S_a < \tau_{ap}$  then
10:       $\Gamma^{(t)} \leftarrow \Gamma^{(t)} \cup \{\text{CRITIQUE}(x, a, d_a)\}$ 
11:   end if
12: end for
13: if  $\Gamma^{(t)} = \emptyset$  then
14:   break
15: end if
16:  $C^{(t+1)} \leftarrow \text{REVISE}(C^{(t)}, \Gamma^{(t)})$ ;  $t \leftarrow t + 1$ 
17: until  $C^{(t)} = C^{(t-1)}$  or  $t = T_{ap}$ 
18:  $C^* \leftarrow \{a \mid d_a \in \mathcal{D}^{(t)}, \text{mentioned}_a = 1\}$ 
19:  $\mathcal{O}_{ap} \leftarrow \{(a, S_a, E_a)\}_{a \in C^*}$ 
20: return  $C^*, \mathcal{O}_{ap}$ 

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where $y_a^{(t)}$ denotes the proposed sentiment label, $r_a^{(t)}$ is the LLM confidence score, and $E_a^{(t)} = \{e_1, \dots, e_m\}$ is a set of extracted evidence spans, each evidence span e must be an exact excerpt from the input comment x similar to the extracting strategy in AP. Sentiment expressions related to the same aspect often recur across student responses, even when expressed with different wording. Therefore, cross-instance verification can help improve consistency and reliability. To support this, SP maintains a similar memory bank to the one used in AP to store previously validated aspect-sentiment evidence patterns and their validation scores:

$$\mathcal{M}_{sp} = \{(a_i, y_i, E_i, \gamma_i)\}_{i=1}^N,$$

where a_i denotes an aspect, $y_i \in \mathcal{Y}$ is the stored sentiment label, E_i is the associated evidence phrase set, and γ_i is the validation score of that memory entry. SP memory is populated before evaluation by running the proposer-validator over each annotated aspect in the training split and storing only predictions with $S_a \geq \tau_{sp}$, which is then frozen for development and test.

The **Sentiment Validator** evaluates whether the proposed sentiment label is well supported by the evidence set E_a . The validator combines two complementary signals. The first is a **memory-based similarity signal**, which compares the proposed evidence against memorized evidence patterns of the same aspect. For each embedded evidence span q , the validator retrieves the top- K most similar memory entries and accumulates similarity support for each sentiment label in $\mathcal{Y}$.

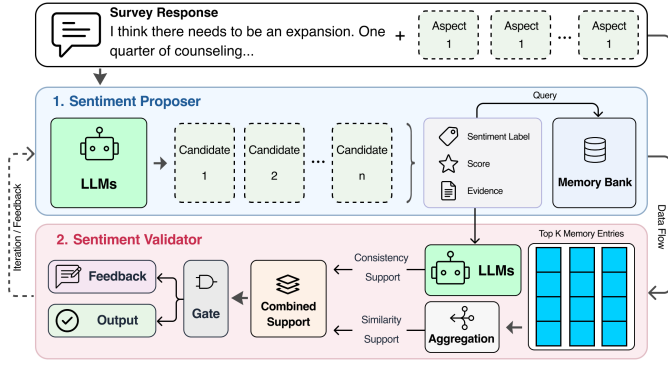


Fig. 3. Overview of the Sentiment Predictor module, where an LLM-based proposer generates aspect-conditioned sentiment predictions with evidence, and a memory-augmented validator refines decisions via similarity-based aggregation, consistency checking, and gated validation.

For each sentiment label y , if we denote the cosine similarity between the query span q and a memory entry $m \in \mathcal{M}_{\text{sp}}$ by $\text{sim}(q, m)$, the support score for label y is aggregated as:

$$\text{score}_{\text{mem}}(y) = \frac{1}{|\mathcal{N}_y| + \epsilon} \sum_{m \in \mathcal{N}_y} \text{sim}(q, m) \cdot S_{\text{rel}}(q, a),$$

where $\mathcal{N}_y$ is the set of retrieved memory entries with label y , and $S_{\text{rel}}(q, a)$ measures how relevant the evidence phrase is to the description of the target aspect by calculating the cosine similarity between q and a . The second signal is an **LLM-based sentiment consistency signal**. Instead of relying only on retrieved memory, the validator also uses an LLM to estimate the sentiment distribution expressed by the evidence phrase itself with respect to aspect a . This yields a probability vector:

$$\mathbf{p}_{\text{llm}}(q, a) = (p_{\text{pos}}, p_{\text{neg}}, p_{\text{neu}}),$$

where the three probabilities sum approximately to one. For each label y , the model computes a margin-based support score relative to the strongest competing label:

$$\text{supp}_{\text{llm}}(y) = \text{clip}\left(p_y - \max_{y' \neq y} p_{y'}, -1, 1\right).$$

Similarly, the memory-based signal $\text{score}_{\text{mem}}(y)$ is also converted into a label-level margin score by comparing the support of the proposed label against the best alternative, denoted as $\text{supp}_{\text{sim}}(y)$. Therefore, both signals are expressed on a common scale and can be fused consistently. The final validation support combines both signals:

$$\text{supp}(y) = \alpha \cdot \text{supp}_{\text{sim}}(y) + (1 - \alpha) \cdot \text{supp}_{\text{llm}}(y),$$

where $\alpha \in [0, 1]$ is the hyperparameter that controls the balance between retrieval-based similarity support and the LLM-based sentiment consistency signal; we set $\alpha = 0.6$ in all experiments. For the proposed label $y_a^{(t)}$, the final validation score is mapped into $[0, 1]$ as

$$S_a^{(t)} = \frac{\text{supp}(y_a^{(t)}) + 1}{2}.$$

Algorithm 2 Sentiment Predictor

Require: Text x ; detected aspects C^* ; label space $\mathcal{Y}$; memory bank $\mathcal{M}_{\text{sp}}$; threshold τ_{sp} ; max iterations T_{sp}

Ensure: Aspect-level sentiment output $\mathcal{O}_{\text{sp}}$

```

1:  $\mathcal{O}_{\text{sp}} \leftarrow \emptyset$ 
2: for all  $a \in C^*$  do
3:    $\mathcal{F} \leftarrow \emptyset, o_a \leftarrow \emptyset$ 
4:   for  $t = 1$  to  $T_{\text{sp}}$  do
5:      $p_a^{(t)} = (y_a^{(t)}, r_a^{(t)}, E_a^{(t)}) \leftarrow \text{SENTIMENTPROPOSER}(x, a, \mathcal{F})$ 
6:      $\mathcal{M}_a \leftarrow \{m \in \mathcal{M}_{\text{sp}} \mid m.\text{aspect} = a\}$ 
7:      $S_a^{(t)}, \Gamma_a^{(t)} \leftarrow \text{SENTIMENTVALIDATOR}(x, a, p_a^{(t)}, \mathcal{M}_a)$ 
8:     if  $o_a = \emptyset$  or  $S_a^{(t)} > \hat{S}_a$  then
9:        $o_a \leftarrow (y_a^{(t)}, E_a^{(t)}, S_a^{(t)})$ 
10:    end if
11:    if  $S_a^{(t)} \geq \tau_{\text{sp}}$  then
12:      if building memory: update  $\mathcal{M}_{\text{sp}}$ ; break
13:    end if
14:     $\mathcal{F} \leftarrow \mathcal{F} \cup \{\Gamma_a^{(t)}\}$ 
15:  end for
16:   $\mathcal{O}_{\text{sp}} \leftarrow \mathcal{O}_{\text{sp}} \cup \{(a, o_a)\}$ 
17: end for
18: return  $\mathcal{O}_{\text{sp}}$ 

```

A candidate is accepted when $S_a^{(t)} \geq \tau_{\text{sp}}$; otherwise, the validator feedback guides revision of its evidence or label. Refinement stops upon acceptance or after T_{sp} rounds, retaining the highest-scoring candidate as uncertain if none passes. The final output is:

$$o_a = (\hat{y}_a, \hat{E}_a, \hat{S}_a),$$

where $\hat{y}_a$, $\hat{E}_a$, and $\hat{S}_a$ are the final label, evidence, and score. Accepted predictions update $\mathcal{M}_{\text{sp}}$ only during training-memory construction; updates are disabled during evaluation.

V. EXPERIMENTS

In this section, we evaluate the proposed ARISE framework on the AWARE-SMILE dataset and conduct extensive studies to analyze the effectiveness of the proposed AP and SP modules. We investigate the following research questions:

- **RQ1:** How does ARISE compare with traditional machine learning models, transformer-based baselines, task-specific ABSA frameworks, and direct LLM prompting methods for aspect prediction and sentiment prediction?
- **RQ2:** Does the proposed proposer-validator framework improve retrieval-augmented inference across different LLM backbones without parameter fine-tuning?
- **RQ3:** How do the proposed AP and SP modules affect performance across different aspect and model components?
- **RQ4:** How effective are retrieval-augmented validation and iterative refinement mechanisms in improving prediction reliability and reasoning consistency?

A. Experimental Settings

1) **Dataset and Evaluation Metrics:** The dataset is divided once into fixed training, development, and test sets (75%/12.5%/12.5%) shared by all methods. Only training samples construct $\mathcal{M}_{ap}$ and $\mathcal{M}_{sp}$, which are both frozen before evaluation. Additionally, we evaluated the performance of LLM-based methods on the full AWARE-SMILE dataset.

We report Precision, Recall, Accuracy, and F1-score. For aspect prediction, evaluation follows a multi-label setting. For sentiment prediction, evaluation is performed conditioned on gold aspects. Specifically, sentiment evaluation is conducted only on aspects that appear in the gold annotations, following a gold-aspect conditional evaluation protocol. Under this protocol, sentiment precision, recall, and F1 are micro-averaged; they coincide when every gold aspect receives one valid label, whereas missing or UNSURE predictions are counted as false negatives.

2) **Baseline Methods:** We compare ARISE against four categories of baselines, including traditional machine learning models, fine-tuned transformer models, task-specific ABSA frameworks, and direct LLM-based inference methods.

- **Traditional ML Models:** Logistic Regression, Support Vector Machine, and XGBoost with TF-IDF features.
- **Transformer-based Models:** Fine-tuned RoBERTa [6] and DeBERTa [7] models for aspect prediction and aspect-conditioned sentiment classification.
- **Task-specific ABSA Models:** We include three representative ABSA models with different structured sentiment reasoning strategies. BERT-ASC [33] performs aspect prediction using auxiliary-sentence grounding and semantic indicator matching. ChatABSA [34] adopts instruction-guided structured sentiment extraction with constrained prompting and formatted outputs. Syn-Chain [35] performs syntax-aware multi-step sentiment reasoning through dependency-guided opinion extraction and sentiment inference.
- **Direct LLM-based Baselines:** Prompt-based GPT-4o mini [36], Claude Haiku 4.5 [37], and Gemini 2.5 Flash [38] models that directly generate aspect and sentiment predictions in a zero-shot setting without retrieval augmentation or iterative validation.

3) **Implementation Details:** Traditional machine learning baselines (LR, SVM, and XGBoost) are implemented using TF-IDF features, while transformer-based baselines use fine-tuned RoBERTa and DeBERTa models under the same training/dev/test split with gold-aspect-conditioned sentiment prediction. Task-specific ABSA baselines follow their original implementations, where BERT-ASC is adapted to the AWARE-SMILE domain using semantically related keywords and evidence spans for auxiliary-sentence grounding, while ChatABSA and Syn-Chain retain their original prompting and reasoning strategies. ARISE is implemented following the method introduced in the framework section. All LLM modules use deterministic decoding ($temperature = 0$) with schema-constrained prompting. The AP settings are

TABLE III

ASPECT & SENTIMENT INFERENCE PERFORMANCE ON TEST SET. UPARROW VALUES INDICATE IMPROVEMENTS AFTER INCORPORATING ARISE ON LLMs. BOLD AND UNDERLINED VALUES DENOTE THE BEST AND SECOND-BEST RESULTS.

Model	Aspect Prediction			Sentiment Prediction		
	P	R	F1	P	R	F1
LR	<u>0.574</u>	0.387	0.462	0.744	0.339	0.490
SVM	0.607	0.376	0.464	0.766	0.345	0.476
XGBoost	0.583	0.322	0.415	0.720	0.720	0.720
RoBERTa	0.388	0.552	0.456	0.716	0.398	0.511
DeBERTa	0.343	0.713	0.463	0.817	0.817	0.817
BERT-ASC	0.392	0.437	0.413	<u>0.842</u>	0.368	0.512
ChatABSA	0.500	0.644	0.563	0.841	0.707	0.768
Syn-Chain	Not supported			0.880	0.537	0.667
GPT	0.411	0.713	0.521	0.609	0.609	0.609
Claude	0.459	0.602	0.521	0.627	0.627	0.627
Gemini	0.562	0.497	0.528	0.643	0.643	0.643
ARISE_{GPT}	0.472	0.735	<u>0.575</u> ↑	0.711	0.627	0.667↑
ARISE_{Claude}	0.440	<u>0.746</u>	<u>0.553</u> ↑	0.787	0.665	0.721↑
ARISE_{Gemini}	0.467	0.862	0.606 ↑	0.804	<u>0.759</u>	<u>0.781</u> ↑

TABLE IV

ASPECT & SENTIMENT INFERENCE PERFORMANCE ON FULL DATASET. UPARROW VALUES INDICATE IMPROVEMENTS AFTER INCORPORATING ARISE ON LLMs. BOLD AND UNDERLINED VALUES DENOTE THE BEST AND SECOND-BEST RESULTS.

Model	Aspect Prediction			Sentiment Prediction		
	P	R	F1	Mi. F1	Ma. F1	ACC
GPT	0.421	0.713	0.530	0.620	0.426	0.620
Claude	0.412	0.744	0.530	0.688	0.467	0.688
Gemini	0.574	0.520	0.546	0.654	0.404	0.654
ARISE_{GPT}	0.464	0.708	<u>0.561</u> ↑	0.734	0.653	0.691↑
ARISE_{Claude}	0.435	<u>0.749</u>	<u>0.551</u> ↑	<u>0.816</u>	<u>0.703</u>	<u>0.756</u> ↑
ARISE_{Gemini}	<u>0.486</u>	0.880	0.626 ↑	0.821	0.776	0.798 ↑

$(K, M, N, T_{ap}) = (8, 5, 3, 2)$ with $\tau_{ap} = 0.5$ for GPT and 0.4 for Claude/Gemini. The SP settings are $(K_m, \alpha, \tau_{sp}, T_{sp}) = (12, 0.6, 0.5, 3)$.

B. Results and Analysis

1) **Main Results on the Test Set:** Table III reports the overall comparison between ARISE and multiple baseline methods on the held-out test set. Traditional machine learning models achieve relatively high precision but generally suffer from low recall, indicating limited ability to capture implicit and context-dependent aspect expressions in open-ended student feedback. Fine-tuned transformer models remain highly competitive under supervised settings, with DeBERTa achieving the strongest sentiment prediction performance. Among task-specific ABSA baselines, ChatABSA shows strong overall structured prediction capability, while Syn-Chain achieves particularly high sentiment precision, suggesting that syntax-aware reasoning helps reduce noisy sentiment predictions. In contrast, BERT-ASC performs less effectively on the AWARE-SMILE domain, indicating that auxiliary-sentence grounding methods designed for restaurant-review benchmarks may generalize poorly to abstract mental-health service aspects.

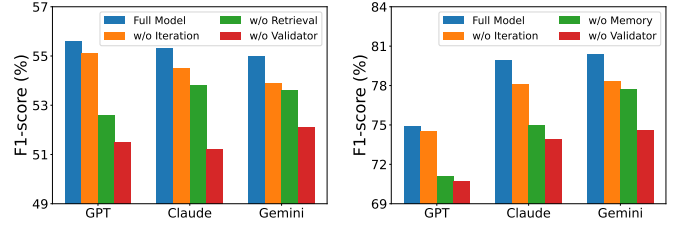
TABLE V
PER-ASPECT ASPECT PREDICTION (TOP) AND SENTIMENT PREDICTION (BOTTOM) PERFORMANCE ACROSS EIGHT ASPECT CATEGORIES. BOLD AND UNDERLINED VALUES DENOTE THE BEST AND SECOND-BEST RESULTS.

Model	Aspect 1			Aspect 2			Aspect 3			Aspect 4			Aspect 5			Aspect 6			Aspect 7			Aspect 8		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Per-aspect ASPECT prediction performance across eight aspect categories																								
GPT	0.21	0.15	0.18	0.61	0.91	<u>0.73</u>	0.28	<u>0.90</u>	0.43	0.29	0.72	0.41	0.44	0.81	0.57	0.63	0.54	0.58	<u>0.64</u>	0.78	0.71	0.13	0.46	0.21
Claude	0.19	0.51	<u>0.28</u>	0.49	0.97	0.65	0.32	0.94	0.48	0.48	0.72	<u>0.58</u>	0.41	0.79	0.54	0.68	0.68	0.68	<u>0.55</u>	0.93	0.69	0.26	0.61	<u>0.36</u>
Gemini	0.18	0.08	0.11	0.72	0.45	0.55	0.40	0.52	0.45	0.63	0.30	0.40	0.49	0.32	0.39	1.00	0.32	0.49	0.78	0.33	0.46	0.20	0.22	0.21
ARISE_{GPT}	<u>0.22</u>	0.30	0.25	0.62	0.83	0.71	0.31	0.81	0.45	0.51	0.63	0.56	0.52	0.68	0.59	0.78	0.50	0.61	0.55	0.86	0.67	0.30	0.44	0.35
ARISE_{Claude}	0.18	<u>0.56</u>	0.27	<u>0.65</u>	0.77	0.71	<u>0.32</u>	0.82	<u>0.46</u>	<u>0.53</u>	<u>0.75</u>	0.62	<u>0.51</u>	0.65	<u>0.57</u>	<u>0.86</u>	0.43	0.57	0.54	<u>0.95</u>	0.69	0.39	0.25	0.31
ARISE_{Gemini}	0.27	0.73	0.39	0.63	<u>0.95</u>	0.75	0.29	0.84	0.43	0.25	0.79	0.38	0.22	0.81	0.35	0.65	<u>0.59</u>	<u>0.62</u>	0.55	0.97	<u>0.70</u>	0.39	<u>0.58</u>	0.47
Per-aspect SENTIMENT prediction performance across eight aspect categories																								
GPT	0.51	0.35	0.34	0.56	0.39	0.44	0.50	0.34	0.41	0.46	0.31	0.36	0.49	0.34	0.40	0.29	0.10	0.12	0.59	0.42	0.48	0.56	0.37	0.44
Claude	<u>0.50</u>	0.37	0.36	<u>0.58</u>	0.44	0.48	<u>0.57</u>	0.43	0.48	0.51	0.44	0.43	<u>0.53</u>	0.37	0.43	<u>0.44</u>	0.30	0.34	0.61	0.48	0.52	0.55	0.41	0.47
Gemini	0.49	0.32	0.32	0.58	0.39	0.44	0.56	0.36	0.43	<u>0.52</u>	0.29	0.35	0.56	0.32	0.40	0.44	<u>0.31</u>	<u>0.34</u>	0.56	0.40	0.46	0.60	0.40	<u>0.48</u>
ARISE_{GPT}	0.50	0.38	0.39	0.57	0.47	0.51	0.50	0.41	0.45	0.44	0.35	0.39	0.49	<u>0.43</u>	<u>0.46</u>	0.37	0.22	0.28	0.45	0.48	0.46	0.52	0.38	0.44
ARISE_{Claude}	0.46	0.39	<u>0.42</u>	0.53	<u>0.50</u>	<u>0.52</u>	0.50	0.42	0.45	0.49	<u>0.42</u>	<u>0.43</u>	0.50	0.41	0.45	0.40	0.36	0.37	0.55	<u>0.53</u>	<u>0.54</u>	<u>0.58</u>	0.45	0.50
ARISE_{Gemini}	0.46	0.42	0.44	0.63	0.56	0.58	0.60	0.48	0.53	0.56	0.44	0.49	0.50	0.56	0.53	0.38	0.21	0.27	<u>0.60</u>	0.59	0.59	0.29	0.23	0.26

Direct LLM baselines demonstrate stronger semantic understanding and aspect coverage than traditional supervised models, but their predictions remain unstable without explicit verification. After incorporating ARISE, all three LLM backbones consistently improve in both aspect prediction and sentiment prediction. In particular, retrieval-guided validation and iterative refinement substantially improve aspect recall and overall aspect F1, with Gemini+ARISE achieving the best overall aspect performance. Similar improvements are observed in sentiment prediction, where the proposer–validator paradigm and memory-augmented validation produce more reliable and consistent aspect-level sentiment decisions across all backbone models. Overall, the results suggest that retrieval-augmented validation can improve fine-grained ABSA performance under the adopted evaluation protocol.

2) *Main Results on the Full AWARE-SMILE Dataset:* Table IV evaluates the retrieval-augmented inference performance of ARISE on the full AWARE-SMILE without parameter fine-tuning. Overall, ARISE consistently improves aspect prediction across all LLM backbones, particularly in recall and overall F1, indicating that iterative refinement and retrieval-guided validation help recover implicit or missing aspect candidates. Among all models, Gemini+ARISE achieves the strongest aspect prediction performance with the highest recall and F1 score.

For sentiment prediction, ARISE introduces improvements across all backbone models. Compared with direct prompting baselines, the proposed SP module substantially improves Micro-F1, Macro-F1, and Accuracy, demonstrating more reliable and balanced sentiment reasoning under the adopted retrieval-augmented setting. Gemini+ARISE again achieves the best overall sentiment performance. The consistent gains across OpenAI, Anthropic, and Google backbones suggest that the proposed proposer–validator framework can improve



(a) Aspect Predictor (AP) (b) Sentiment Predictor (SP)
Fig. 4. Ablation study of AP and SP across different LLM backbones.

prediction performance across the three evaluated backbones.

3) *Per-Aspect Performance Analysis:* Table V reports the per-aspect performance for both aspect prediction and sentiment prediction. Overall, ARISE improves performance on most aspect categories across all LLM backbones, especially for aspect prediction. The gains are noticeable on Aspect 1, Aspect 2, Aspect 6, Aspect 7, and Aspect 8, suggesting that iterative validation and retrieval-guided refinement help recover implicit or semantically ambiguous aspect mentions.

The results also show substantial variation across aspect categories. Aspect 2, Aspect 6, and Aspect 7 consistently achieve stronger performance across models, indicating clearer semantic boundaries and lexical cues, while Aspect 1 and Aspect 8 remain more challenging due to broader or overlapping semantics. Compared with aspect prediction, sentiment prediction exhibits more stable improvements across categories, demonstrating that the SP module effectively enhances aspect-level sentiment consistency through memory-augmented validation and iterative refinement.

4) *Ablation Study:* Figure 4 presents the ablation results of AP and SP across different LLM backbones. Removing iterative refinement causes moderate degradation, while removing retrieval or memory modules leads to larger performance drops, indicating that external prototype evidence

TABLE VI
CASE STUDY OF THE PROPOSED ASPECT PREDICTOR FRAMEWORK IN ARISE.

Input	Text Response: "I have looked at group counseling but they are still remote. I am tired of remote and don't want it simple because it's remote. if it were in person I would definitely go to group therapy." Gold Label: "Counseling Service"
Proposer Prediction 1 Memory Support	Aspect: "Counseling Service" Evidence: "group counseling" Positive: "one on one counseling was working well.", "we have a varieties of counseling,"; Negative: "Service Availability: need more scheduled communication, perhaps group sessions", "Mental Health Service: there are definitely known resources like counselors and group therapy"
Validator Score	$mentioned_a = \text{True}$, $S_a = 1.0$
Proposer Prediction 2 Memory Support	Aspect: "Service Availability" Evidence: "they are still remote", "i am tired of remote" Positive: "i think there needs to be more options for online students.", "they are always full and i am unable to ever book a session when i need it most."; Negative: "General: i am a fully online student and i wish i was able to use more telehealth services", "Counseling Service: flu should focus more on online counseling because the pandemic forces students to stay home"
Validator Score	$mentioned_a = \text{True}$, $S_a = 0.3$
Critiques	$why_wrong =$ "Service Availability: The text expresses a preference for in-person services rather than discussing the availability of services. The focus is on the dissatisfaction with remote options rather than the availability of services themselves", $sugg_drop =$ "Service Availability"
Proposer Prediction 1 Memory Support	Aspect: "Counseling Service" Evidence: "group counseling" Positive: "one on one counseling was working well.", "we have a varieties of counseling,"; Negative: "Service Availability: need more scheduled communication, perhaps group sessions", "Mental Health Service: there are definitely known resources like counselors and group therapy"
Validator Score	$mentioned_a = \text{True}$, $S_a = 1.0$
Output	Aspect(s): "Counseling Service"

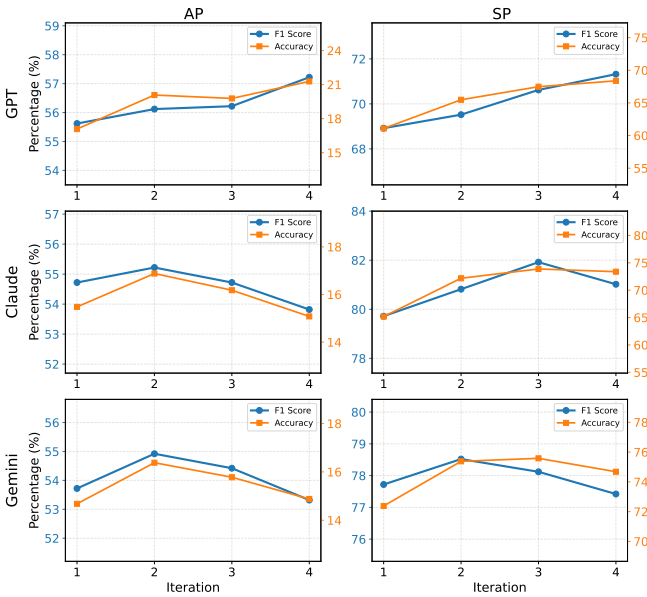


Fig. 5. Sensitivity analysis of the number of iterations for both AP and SP using three LLM backbones. Accuracy in APs denotes exact-match accuracy under the multi-label setting, requiring the complete predicted aspect set to match the gold set.

and historical sentiment patterns provide important validation signals. The most significant decline occurs when the validator is removed entirely in both AP and SP, showing that direct LLM generation alone is insufficient for reliable aspect and sentiment prediction. The overall trends remain highly consistent across OpenAI, Anthropic, and Google backbones, suggesting that the validator is an important component across the three evaluated backbones.

5) Effect of Iterative Refinement: Figure 5 illustrates the performance variation under different iteration numbers for AP and SP across three LLM backbones. Overall, both AP and SP benefit from iterative refinement during early iterations, while excessive iterations yield diminishing returns or slight degradation. For AP, the best performance is typically achieved at the second iteration, indicating that retrieval-guided critique effectively improves aspect discovery in early

refinement stages. However, excessive refinement may introduce over-correction and semantic drift. Compared with AP, SP exhibits smoother and more stable improvement curves. Performance generally increases during early iterations and stabilizes around iteration 3 and 4 across all backbone models. This suggests that memory-augmented validation and iterative evidence refinement effectively improve sentiment consistency once aspect evidence has been localized. Overall, the results demonstrate that a certain number of refinement rounds is necessary for both AP and SP, validating the effectiveness of the iterative proposer-validator design.

6) Case Study: As shown in Table VI, the proposer initially predicts two candidate aspects from the input text: Counseling Service and Service Availability, together with their corresponding evidence spans. For each candidate, the validator retrieves positive prototypes from the same aspect category and hard negatives from semantically similar categories using the AP memory bank, then computes $mentioned_a$ and validation score S_a based on the retrieved evidence and grounding quality of the extracted spans. Among the two candidates, Service Availability is rejected as its validation score falls below the threshold ($S_a = 0.3$). The validator further generates critique feedback indicating that the text mainly expresses dissatisfaction with remote counseling formats and preference for in-person therapy, rather than discussing service accessibility itself. In the second iteration, the proposer removes Service Availability while retaining Counseling Service; all remaining candidates pass validation, so the iterative process terminates and produces the final output. This example demonstrates that retrieval-augmented validation and critique-based refinement effectively reduce semantically ambiguous aspect predictions and improve prediction reliability.

VI. CONCLUSION

In this paper, we propose ARISE, a proposer-validator framework combining retrieval-augmented aspect validation and memory-enhanced sentiment verification without LLM fine-tuning. Experiments show consistent improvements across multiple backbones. Limitations include the modest, single-domain dataset; manual taxonomy refinement; dependence

on proprietary APIs and labeled memory; and gold-aspect-conditioned SP evaluation. Future work will directly evaluate evidence spans and end-to-end prediction, and study transfer to other feedback domains.

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